CISCO ASA FIREWALL

Complete Study Notes · Basic to Advanced

Topics Covered

- ✓ Firewall Basics & Architecture
- ✓ Interface Config & Security Levels
- ✓ NAT — Static, Dynamic, PAT
- ✓ Access Control Lists (ACLs)
- ✓ Routing — Static & Dynamic
- ✓ VPN — Site-to-Site & Remote Access
- ✓ High Availability (Failover)
- ✓ MPF — Modular Policy Framework
- ✓ Logging, Monitoring & Troubleshooting
- ✓ Best Practices & Quick Reference

1 · Firewall Fundamentals & ASA Overview

1.1 What is ASA?

Cisco ASA (Adaptive Security Appliance) is a stateful firewall + VPN device. It tracks every connection in a **state table** and allows return traffic automatically — eliminating manual inbound rules for established sessions.

Property	Details
Firewall Type	Stateful — tracks TCP/UDP session state
OS	Cisco ASA OS (ASA-OS / ASDM)
Key Models	ASA 5505, 5506-X, 5510, 5515-X, 5525-X, 5585-X, Firepower
Management	CLI (SSH/Console), ASDM (GUI), Cisco FMC
Licensing	Base, Security Plus, AnyConnect, IPS

1.2 Security Levels (Core Concept)

KEY RULE: Higher security level → Lower: traffic ALLOWED by default. Lower → Higher: traffic BLOCKED by default (needs ACL).

Security Level	Interface Role
100	Inside (LAN) — fully trusted
50	DMZ — semi-trusted (web/mail servers)
0	Outside (Internet) — untrusted
1–99	Any custom value for additional zones

1.3 ASA Packet Flow

Understanding packet flow is essential for troubleshooting:

- 1. Packet arrives on ingress interface
- 2. Check existing connection table (if match → fast-path, skip steps below)
- 3. Check ACL on ingress interface (inbound)
- 4. Route lookup — find egress interface
- 5. NAT translation (un-NAT on ingress, NAT on egress)
- 6. Check ACL on egress interface (outbound) — rarely used
- 7. Inspect / MPF policy checks
- 8. Transmit on egress interface

2 · Interface Configuration

2.1 Basic Interface Setup

```
! Enter privileged exec mode
ASA# enable
ASA# configure terminal

! Configure Inside interface
ASA(config)# interface GigabitEthernet0/0
ASA(config-if)# nameif inside
ASA(config-if)# security-level 100
ASA(config-if)# ip address 192.168.1.1 255.255.255.0
ASA(config-if)# no shutdown

! Configure Outside interface
ASA(config)# interface GigabitEthernet0/1
ASA(config-if)# nameif outside
ASA(config-if)# security-level 0
ASA(config-if)# ip address 203.0.113.1 255.255.255.0
ASA(config-if)# no shutdown

! Configure DMZ interface
ASA(config)# interface GigabitEthernet0/2
ASA(config-if)# nameif dmz
ASA(config-if)# security-level 50
ASA(config-if)# ip address 10.10.10.1 255.255.255.0
ASA(config-if)# no shutdown
```

2.2 Sub-interfaces (VLANs)

```
ASA(config)# interface GigabitEthernet0/0.10
ASA(config-subif)# vlan 10
ASA(config-subif)# nameif VLAN10
ASA(config-subif)# security-level 80
ASA(config-subif)# ip address 10.10.10.1 255.255.255.0
```

2.3 Verification Commands

Command	Purpose
show interface ip brief	Quick view of all interfaces and IPs
show nameif	Interface names and security levels
show ip address	Interface IP assignments
show interface GigX	Detailed stats (errors, drops)

3 · Network Address Translation (NAT)

REMEMBER: ASA 8.3+ uses OBJECT-BASED NAT. Pre-8.3 syntax (nat/global commands) is deprecated.

3.1 NAT Types

NAT Type	Use Case
Static NAT	1-to-1 mapping. Used for servers needing fixed public IP
Dynamic NAT	Pool of IPs — many-to-many, no port translation
PAT (Masquerade)	Many-to-one using ports. Most common for LAN → Internet
Identity NAT	Translate to same IP (used in VPN split tunnel)
Twice NAT	Translates both source AND destination simultaneously

3.2 Static NAT — Server Publishing

```
! Publish DMZ web server (10.10.10.10) as public 203.0.113.10
ASA(config)# object network WEB-SERVER
ASA(config-network-object)# host 10.10.10.10
ASA(config-network-object)# nat (dmz,outside) static 203.0.113.10
```

3.3 PAT — Inside Users to Internet

```
! PAT using the outside interface IP (most common)
ASA(config)# object network LAN-USERS
ASA(config-network-object)# subnet 192.168.1.0 255.255.255.0
ASA(config-network-object)# nat (inside,outside) dynamic interface

! PAT using a specific public IP
ASA(config)# object network LAN-USERS
ASA(config-network-object)# nat (inside,outside) dynamic 203.0.113.5
```

3.4 Static Port Forwarding (PAT Port Redirect)

```
! Forward TCP 443 on public IP to internal server
ASA(config)# object network HTTPS-SERVER
ASA(config-network-object)# host 192.168.1.100
ASA(config-network-object)# nat (inside,outside) static 203.0.113.1 service tcp 443 443
```

3.5 NAT Verification

Command	Purpose
show nat	All NAT policies and hit counts
show nat detail	Detailed NAT translation info
show xlate	Active NAT translation table
show conn	Active connection table (also shows NAT)
clear xlate	Clear all NAT translations (use carefully!)
packet-tracer	Simulate packet and trace NAT+ACL decisions

4 · Access Control Lists (ACLs)

4.1 ACL Fundamentals

ACLs control traffic INBOUND to an interface. Key rules:

- ASA adds implicit DENY ALL at end of every ACL
- ACLs are processed top-down — first match wins
- For return traffic of existing sessions, the state table bypasses ACL checks
- Apply ACL inbound on the interface where traffic enters ASA

4.2 Extended ACL — Syntax

```
! Syntax: access-list NAME extended {permit|deny} PROTO SRC DST

! Allow HTTP from DMZ to outside
ASA(config)# access-list OUTSIDE-IN extended permit tcp any host 203.0.113.10 eq 80
ASA(config)# access-list OUTSIDE-IN extended permit tcp any host 203.0.113.10 eq 443

! Allow ICMP from inside to anywhere
ASA(config)# access-list INSIDE-OUT extended permit icmp 192.168.1.0 255.255.255.0 any

! Deny a specific host
ASA(config)# access-list INSIDE-OUT extended deny ip host 192.168.1.50 any

! Apply ACL to interface (inbound direction)
ASA(config)# access-group OUTSIDE-IN in interface outside
ASA(config)# access-group INSIDE-OUT in interface inside
```

4.3 Object-Groups (Cleaner ACLs)

```
! Define service group
ASA(config)# object-group service WEB-PORTS tcp
ASA(config-service-object-group)# port-object eq 80
ASA(config-service-object-group)# port-object eq 443
ASA(config-service-object-group)# port-object eq 8080

! Define network group
ASA(config)# object-group network TRUSTED-HOSTS
ASA(config-network-object-group)# network-object host 192.168.1.10
ASA(config-network-object-group)# network-object host 192.168.1.11

! Use groups in ACL — much cleaner!
ASA(config)# access-list INSIDE-IN extended permit tcp object-group TRUSTED-HOSTS any
object-group WEB-PORTS
```

4.4 ACL Best Practices

Practice	Details
Order matters	Put most specific rules FIRST, general rules last
Use remarks	access-list NAME remark DESCRIPTION — for documentation
Object groups	Reduces ACL size and simplifies management
Logging	Add 'log' at end of ACE to log matches
Time-based ACLs	time-range command restricts ACL to schedules

Avoid outbound ACLs	Only needed in special cases; inbound is standard

4.5 ACL Verification

Command	Purpose
show access-list	List all ACLs with hit counts
show access-group	Show which ACLs are applied to interfaces
packet-tracer input inside tcp 192.168.1.1 1234 8.8.8.8 80	Simulate and trace packet
clear access-list NAME counters	Reset hit counters

5 · Routing

5.1 Static Routing

```
! Default route (most important — points to ISP)
ASA(config)# route outside 0.0.0.0 0.0.0.0 203.0.113.254 1

! Static route to internal subnet via inside router
ASA(config)# route inside 10.20.0.0 255.255.0.0 192.168.1.254 1

! Floating static route (backup, higher metric)
ASA(config)# route outside2 0.0.0.0 0.0.0.0 203.0.114.254 254
```

5.2 Dynamic Routing — OSPF

```
! Enable OSPF process 1
ASA(config)# router ospf 1
ASA(config-router)# router-id 1.1.1.1
ASA(config-router)# network 192.168.1.0 255.255.255.0 area 0
ASA(config-router)# network 10.10.10.0 255.255.255.0 area 0
ASA(config-router)# default-information originate always
ASA(config-router)# log-adj-changes
```

5.3 Dynamic Routing — EIGRP

```
ASA(config)# router eigrp 100
ASA(config-router)# network 192.168.1.0
ASA(config-router)# network 10.10.10.0
ASA(config-router)# no auto-summary
```

Command	Purpose
show route	Routing table — check all entries
show ospf neighbor	OSPF neighbor adjacencies
show ospf database	OSPF LSDB entries
show eigrp neighbors	EIGRP neighbors
ping / traceroute	Test reachability

6 · VPN Configuration

6.1 Site-to-Site IKEv1 VPN

Site-to-Site VPN connects two networks over the Internet using IPSec. Two phases:

- Phase 1 (ISAKMP): Authenticate peers, negotiate encryption for Phase 2
- Phase 2 (IPSec): Negotiate encryption for actual data traffic

```
! ── Phase 1: ISAKMP Policy ──────────────────────────────
ASA(config)# crypto isakmp policy 10
ASA(config-isakmp-policy)# authentication pre-share
ASA(config-isakmp-policy)# encryption aes-256
ASA(config-isakmp-policy)# hash sha256
ASA(config-isakmp-policy)# group 14
ASA(config-isakmp-policy)# lifetime 86400

! Enable ISAKMP on outside interface
ASA(config)# crypto isakmp enable outside

! Pre-shared key for peer
ASA(config)# tunnel-group 203.0.114.1 type ipsec-l2l
ASA(config)# tunnel-group 203.0.114.1 ipsec-attributes
ASA(config-tunnel-ipsec)# ikev1 pre-shared-key Sup3rS3cr3t!

! ── Phase 2: IPSec Transform Set ────────────────────────
ASA(config)# crypto ipsec ikev1 transform-set TS esp-aes-256 esp-sha-hmac

! ── Crypto Map ──────────────────────────────────────────
ASA(config)# crypto map VPNMAP 10 match address VPN-TRAFFIC
ASA(config)# crypto map VPNMAP 10 set peer 203.0.114.1
ASA(config)# crypto map VPNMAP 10 set ikev1 transform-set TS
ASA(config)# crypto map VPNMAP interface outside

! ── Interesting Traffic ACL ─────────────────────────────
ASA(config)# access-list VPN-TRAFFIC extended permit ip 192.168.1.0 255.255.255.0
10.2.0.0 255.255.0.0

! ── Identity NAT (bypass NAT for VPN traffic) ───────────
ASA(config)# nat (inside,outside) source static INSIDE-NET INSIDE-NET destination static
REMOTE-NET REMOTE-NET
```

6.2 Remote Access VPN — AnyConnect SSL

```
! Enable WebVPN on outside interface
ASA(config)# webvpn
ASA(config-webvpn)# enable outside
ASA(config-webvpn)# anyconnect image disk0:/anyconnect.pkg 1
ASA(config-webvpn)# anyconnect enable

! IP Pool for VPN clients
ASA(config)# ip local pool VPN-POOL 10.100.0.10-10.100.0.100 mask 255.255.255.0

! Group Policy
ASA(config)# group-policy VPN-POLICY internal
ASA(config)# group-policy VPN-POLICY attributes
ASA(config-group-policy)# vpn-tunnel-protocol ssl-client
ASA(config-group-policy)# split-tunnel-policy tunnelspecified
ASA(config-group-policy)# split-tunnel-network-list value SPLIT-ACL
```

```
ASA(config-group-policy)# dns-server value 8.8.8.8 8.8.4.4

! Tunnel Group
ASA(config)# tunnel-group VPN-USERS type remote-access
ASA(config)# tunnel-group VPN-USERS general-attributes
ASA(config-tunnel-general)# address-pool VPN-POOL
ASA(config-tunnel-general)# default-group-policy VPN-POLICY

! Local user for VPN auth
ASA(config)# username vpnuser password Passw0rd! privilege 0
```

6.3 VPN Verification

Command	Purpose
show crypto isakmp sa	Phase 1 SA status (MM_ACTIVE = UP)
show crypto ipsec sa	Phase 2 SAs + packet counters
show vpn-sessiondb anyconnect	Active AnyConnect sessions
debug crypto isakmp 127	Phase 1 debug (use carefully)
debug crypto ipsec 127	Phase 2 debug
clear crypto ipsec sa peer X.X.X.X	Reset SA for a peer

7 · High Availability — Active/Standby Failover

7.1 Failover Concepts

Type	Description
Active/Standby	One ASA handles traffic; standby mirrors state. On failure, standby takes over.
Active/Active	Both units process traffic (requires multi-context mode)
Stateful Failover	Syncs connection table, NAT xlate, ARP — sessions survive failover
Stateless Failover	Only config syncs — active sessions drop on failover
Failover Link	Dedicated interface for heartbeat + config sync
Stateful Link	Dedicated interface for session state sync (can share with failover link)

7.2 Failover Configuration

```
! ─── On PRIMARY (Active) Unit ──────────────────────────────
ASA(config)# failover lan unit primary
ASA(config)# failover lan interface FAILOVER GigabitEthernet0/3
ASA(config)# failover link STATEFUL GigabitEthernet0/3
ASA(config)# failover interface ip FAILOVER 172.16.0.1 255.255.255.252 standby 172.16.0.2
ASA(config)# failover key MySuperSecret
ASA(config)# failover

! ─── On SECONDARY (Standby) Unit ───────────────────────────
ASA(config)# failover lan unit secondary
ASA(config)# failover lan interface FAILOVER GigabitEthernet0/3
ASA(config)# failover link STATEFUL GigabitEthernet0/3
ASA(config)# failover interface ip FAILOVER 172.16.0.1 255.255.255.252 standby 172.16.0.2
ASA(config)# failover key MySuperSecret
ASA(config)# failover
```

7.3 Failover Verification

Command	Purpose
show failover	Overall status — Active/Standby state
show failover state	Detailed state + last failover reason
failover active	Force failover to this unit
no failover active	Force this unit to standby
write standby	Sync config from active to standby

8 · Modular Policy Framework (MPF)

8.1 MPF Concepts

MPF is ASA's traffic classification and policy engine (similar to Cisco IOS MQC). Three building blocks:

Component	Role
Class Map	Identifies/classifies traffic using match criteria
Policy Map	Assigns actions to classified traffic (inspect, police, drop)
Service Policy	Applies policy map to an interface (or globally)

8.2 Default Global Policy

ASA ships with a default global policy that inspects common protocols:

```
! View the default policy (read-only reference)
ASA# show running-config policy-map

! Default class-map matches all traffic
class-map inspection_default
  match default-inspection-traffic

! Default policy-map applies inspections
policy-map global_policy
  class inspection_default
    inspect dns
    inspect ftp
    inspect http
    inspect icmp
    inspect sip
    inspect skinny

! Applied globally
service-policy global_policy global
```

8.3 Custom MPF — HTTP Inspection + Rate Limit

```
! Step 1: Class-map — match specific traffic
ASA(config)# class-map HTTP-TRAFFIC
ASA(config-cmap)# match port tcp eq 80

! Step 2: Policy-map — assign actions
ASA(config)# policy-map LIMIT-HTTP
ASA(config-pmap)# class HTTP-TRAFFIC
ASA(config-pmap-c)# police input 10000000 conform-action transmit exceed-action drop
ASA(config-pmap-c)# inspect http

! Step 3: Service-policy — apply to interface
ASA(config)# service-policy LIMIT-HTTP interface outside
```

9 · Logging, Monitoring & Troubleshooting

9.1 Syslog Configuration

```
! Enable logging
ASA(config)# logging enable

! Log to ASDM (for GUI monitoring)
ASA(config)# logging asdm informational

! Log to syslog server
ASA(config)# logging host inside 192.168.1.200
ASA(config)# logging trap informational

! Log to console (only for testing)
ASA(config)# logging console errors

! Log to buffer (for CLI review)
ASA(config)# logging buffered informational
ASA(config)# logging buffer-size 1048576
```

9.2 Syslog Severity Levels

Level	Meaning
0 – Emergency	System unusable
1 – Alert	Immediate action needed
2 – Critical	Critical condition
3 – Error	Error condition — use this minimum for production
4 – Warning	Warning — recommend for most environments
5 – Notification	Normal but significant
6 – Informational	Info messages — good for troubleshooting
7 – Debugging	Very verbose — only when debugging specific issue

9.3 Key Troubleshooting Commands

Command	Purpose
packet-tracer input inside tcp 192.168.1.1 1234 8.8.8.8 80	Simulate + trace packet decision
show conn	Active connections — check if session exists
show xlate	NAT translations
show log	View syslog buffer
debug icmp trace	Trace ICMP packets (then 'no debug all')
capture CAP interface outside match ip any any	Packet capture on interface
show capture CAP	View captured packets
copy /pcap capture:CAP tftp://192.168.1.200/cap.pcap	Export capture to Wireshark
show cpu usage	CPU utilization
show memory	Memory usage

show conn count	Number of active connections

9.4 Packet Capture — Full Workflow

```
! 1. Define capture on outside interface
ASA# capture MYCAP interface outside match tcp any host 203.0.113.10

! 2. Reproduce the issue

! 3. View packets
ASA# show capture MYCAP

! 4. Export for Wireshark
ASA# copy /pcap capture:MYCAP tftp://192.168.1.200/out.pcap

! 5. Clean up
ASA# no capture MYCAP
```

10 · Best Practices & Quick Reference

10.1 Security Hardening Checklist

- Change default passwords immediately (enable, ASDM)
- Disable HTTP (ASDM) on outside interface — use SSH + ASDM on inside/mgmt only
- Restrict SSH access to management subnet: ssh 192.168.10.0 255.255.255.0 inside
- Enable AAA authentication: aaa authentication ssh console LOCAL
- Enable NTP for accurate log timestamps: ntp server 192.168.1.1
- Enable service password-encryption
- Set exec timeout: console timeout 5 | ssh timeout 15
- Use IKEv2 over IKEv1 for new VPN deployments
- Apply ACL on outside interface — deny RFC1918 (anti-spoofing)
- Enable Threat Detection: threat-detection basic-threat + threat-detection statistics
- Use object-groups to keep ACLs manageable
- Back up config regularly: copy running-config tftp://server/asa-backup.cfg

10.2 Common Troubleshooting Scenarios

Problem	Solution Approach
Host can't reach Internet	Check route outside, PAT config, ACL hit count, show conn
VPN Phase 1 stuck	Check ISAKMP policy match, pre-shared key, peer IP, show crypto isakmp sa
VPN Phase 2 stuck	Check transform-set match, crypto ACL symmetry (both sides must mirror)
NAT not working	show xlate, verify object + nat statement, check NAT order
ACL blocking unexpectedly	show access-list (hit counts), packet-tracer to identify blocking rule
Failover not syncing	show failover, verify failover link IP, shared key
High CPU	show process cpu-usage, check for scanning/DoS, review logging level
Can't SSH to ASA	Check ssh command, aaa authentication, management ACL, interface reachability

10.3 Most-Used Show Commands — Cheat Sheet

```
! System
show version                    ! ASA version, license, uptime
show running-config             ! Full config
show startup-config             ! Saved config
show clock                      ! Current time

! Interfaces & Routing
show interface ip brief         ! IP + status summary
show route                      ! Routing table
show nameif                     ! Interface names + security levels

! Connections & NAT
show conn count                 ! Total active connections
show conn                       ! Active connection table
show xlate                      ! NAT translations
show nat detail                 ! NAT policy hit counts

! ACL & Policy
```

```
show access-list                ! ACLs with hit counters
show service-policy             ! MPF policy stats

! VPN
show crypto isakmp sa           ! IKEv1 Phase 1
show crypto ikev2 sa            ! IKEv2 SAs
show crypto ipsec sa            ! IPSec SAs + counters
show vpn-sessiondb              ! All active VPN sessions

! HA
show failover                   ! Failover status
write memory                    ! Save config to flash
```

10.4 ASA vs Firepower (FTD) — Quick Comparison

Platform	Use Case
ASA	CLI/ASDM-managed, mature platform, best for traditional L4 firewall + VPN
ASA with FirePOWER Services	ASA + inline IPS/IDS module — best of both worlds
FTD (Firepower Threat Defense)	Next-gen — managed via FMC, deep packet inspection, advanced threat
FDM (Firepower Device Manager)	On-box GUI for FTD, limited to single device management

EXAM TIP: For CCNA Security / CCNP Security: Know packet flow, NAT object syntax (post-8.3), ACL direction (inbound), and VPN Phase 1/2 concepts cold.